

Consider the relation $Z(P, Q, R, S, T)$ having multi-valued dependencies $F = \{P \twoheadrightarrow Q, Q \twoheadrightarrow (S, T)\}$. Which of the following is correct regarding Z and P ?

- ☐ Relation Z is in 4NF and P is a superkey for Z
- ☐ Relation Z is not in 4NF and P is a superkey for Z
- ☐ Relation Z is not in 4NF and P is not a superkey for Z
- ☐ Relation Z is in 4NF and P is not a superkey for Z

Consider the following statements:

2 points

1. If $X \rightarrow Y$, then $X \twoheadrightarrow Y$ is true.
2. If $X \twoheadrightarrow Y$, then $X \rightarrow Y$ is true.
3. According to the **Complementation** rule in MVD, if $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$, then $X \twoheadrightarrow (Z - Y)$.

- ☐ Statements 1 & 2 are correct.
- ☐ Statements 2 & 3 are correct.
- ☐ Only statement 1 is correct
- ☐ All the statements are correct

Choose the correct option which represents the **Augmentation** rule in MVD.

2 points

- ☐ If $X \twoheadrightarrow Y$, then $WX \twoheadrightarrow YZ$
- ☐ If $X \twoheadrightarrow Y$ and $W \supseteq Z$, then $WX \twoheadrightarrow YZ$
- ☐ If $X \twoheadrightarrow Y$, then $X \twoheadrightarrow (R - (X \cup Y))$
- ☐ If $X \twoheadrightarrow Y$ and $W \subseteq Z$, then $WX \twoheadrightarrow YZ$

Choose the correct option which represents the **Transitivity** rule in MVD.

2 points

- ☐ If $X \twoheadrightarrow Y$, then $WX \twoheadrightarrow YZ$
- ☐ If $X \twoheadrightarrow Y$ and $W \supseteq Z$, then $WX \twoheadrightarrow YZ$
- ☐ If $X \twoheadrightarrow Y$, then $X \twoheadrightarrow (R - (X \cup Y))$
- ☐ If $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$, then $X \twoheadrightarrow (Z - Y)$

2 points

Consider relation $\mathbf{R}(X, Y, Z)$ having 4 tuples such that $t_1[X] = t_2[X]$. For a multivalued dependency $X \twoheadrightarrow Y$ to hold true, which of the following conditions must be satisfied?

- ☐ $t_1[X] = t_2[X] = t_3[X] = t_4[X]$
- ☐ $t_3[Y] = t_1[Y]$ and $t_2[Y] = t_4[Y]$
- ☐ $t_4[R - X] = t_1[R - X]$
- ☐ $t_4[X, Z] = t_1[X, Z]$

An MVD $X \twoheadrightarrow Y$ in a relation $\mathbf{R}(X, Y, Z)$ is said to be trivial when,

2 points

- ☐ $X \rightarrow Y$
- ☐ Y is a subset of X
- ☐ $X \cup Y = R$
- ☐ $X \cap Y = R$

2 points

Consider the relation **Student** as shown in Table 4.

Table 4: **Student**

<i>Sname</i>	<i>Course</i>	<i>Instructor</i>	<i>Inst_Room</i>
David	Python	MK Singh	503
David	JAVA	SN Joseph	505
David	Python	SN Joseph	505
David	JAVA	MK Singh	503

Which among the following is an MVD?

- ☐ $Course \twoheadrightarrow \{Inst_Room, Sname\}$
- ☐ $Sname \twoheadrightarrow Course$
- ☐ $Sname \twoheadrightarrow \{Course, Instructor\}$
- ☐ $Sname \twoheadrightarrow Instructor$

2 points

Consider the relation **Student** as shown in Table 5.

Table 5: **Student**

<i>SId</i>	<i>Sname</i>	<i>Course</i>	<i>Instructor</i>	<i>Inst_Room</i>
ME001	David	Python	MK Singh	503
ME001	David	JAVA	SN Joseph	505
ME001	David	Python	SN Joseph	505
ME001	David	JAVA	MK Singh	503

If $Sname \twoheadrightarrow Course$ is an MVD, then which among the following is/are correct?

- ☐ **Student** is in 4NF
- ☐ $\{Sname, SId\} \twoheadrightarrow Course$ is an MVD
- ☐ $SId \twoheadrightarrow \{Instructor, Inst_Room\}$ is an MVD
- ☐ $SId \twoheadrightarrow \{Instructor, Course\}$ is an MVD

Consider the relation schema $\mathbf{R}(A, B)$ having multi-valued dependency $A \twoheadrightarrow B$, where $A \subseteq \mathbf{R}$ and $B \subseteq \mathbf{R}$. For **R** to **2 points** be in 4NF, which of the following conditions must be satisfied?

- ☐ $A \twoheadrightarrow B$ is a non-trivial MVD.
- ☐ $A \twoheadrightarrow B$ is a trivial MVD.
- ☐ A is a super key of **R**.
- ☐ **R** is in 5NF.

Consider the relation $\mathbf{Z}(P, Q, R, S, T)$ having only one functional dependency $\{P \twoheadrightarrow Q\}$, and one multivalued dependency $\{Q \twoheadrightarrow (S, T)\}$. If we decompose **Z** into **Z1**(P, Q) and **Z2**(Q, R, S, T), then which of the following is/are true? **2 points**

- ☐ Relation **Z1** is in 4NF and **P** is a super key for **Z1**
- ☐ Relation **Z2** is not in 4NF and Q is a super key for **Z2**
- ☐ Relation **Z1** is in 4NF and **Z2** is not in 4NF
- ☐ Relation **Z1** is not in 4NF and **Z2** is in 4NF